

GENGMA, China

WMO: 569460

Lat: 23.5273N

Lon: 99.3881E

Elev: 1104

StdP: 88.75

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 4.3	(c) 5.4	(d) 0.8	(e) 4.6	(f) 20.6	(g) 2.2	(h) 5.1	(i) 17.5	(j) 5.0	(k) 22.5	(l) 4.3	(m) 22.5	(n) 0.3	(o) 110	(p) 0.232

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 6.9	(c) 32.1	(d) 19.3	(e) 31.1	(f) 19.6	(g) 30.2	(h) 19.7	(i) 23.1	(j) 27.9	(k) 22.8	(l) 27.1	(m) 22.5	(n) 26.7	(o) 2.3	(p) 270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 22.0	(b) 19.1	(c) 24.3	(d) 21.7	(e) 18.7	(f) 24.1	(g) 21.4	(h) 18.4	(i) 23.9	(j) 75.2	(k) 28.1	(l) 73.9	(m) 27.5	(n) 72.7	(o) 26.9	(p) 25.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.2	3.4	2.7	1.9	33.9	1.1	0.8	1.2	34.5	0.5	35.0	-0.1	35.4	-0.9	36.0
			1.6	23.9	1.0	0.6	0.8	24.4	0.2	24.7	-0.3	25.0	-1.1	25.4
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	20.8	14.2	16.4	19.7	22.4	24.1	24.6	24.2	24.5	24.0	21.8	18.1	14.8	(6)
(7)		DBStd	4.16	1.39	1.79	1.85	1.90	2.06	1.53	1.46	1.30	1.46	1.86	2.00	1.91	(7)
(8)		HDD10.0	1	0	0	0	0	0	0	0	0	0	0	0	1	(8)
(9)		HDD18.3	334	127	58	8	1	1	0	0	0	0	1	29	110	(9)
(10)		CDD10.0	3926	132	179	301	372	437	438	442	449	419	367	242	149	(10)
(11)		CDD18.3	1218	0	3	51	122	179	188	183	191	169	109	21	1	(11)
(12)		CDH23.3	7821	15	152	637	1109	1354	1084	875	1015	896	534	137	13	(12)
(13)		CDH26.7	2047	0	11	147	386	480	277	181	243	219	98	6	0	(13)
(14)	Wind	WSAvg	0.9	0.8	1.1	1.3	1.3	1.1	1.0	0.9	0.9	0.8	0.7	0.6	(14)	
(15)	Precipitation	PrecAvg	1339	20	16	21	54	146	202	295	232	160	129	49	16	(15)
(16)		PrecMax	1792	187	56	78	205	271	334	424	328	284	334	179	101	(16)
(17)		PrecMin	993	0	0	0	5	36	99	156	129	76	36	0	2	(17)
(18)		PrecStd	212	35	19	21	44	71	49	70	47	55	77	48	20	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	28.0	30.6	32.8	33.7	33.2	31.7	31.7	31.2	30.1	27.6	24.7	(19)
MCWB			13.5	14.2	15.4	17.2	19.2	20.7	22.5	22.6	21.7	20.5	17.9	15.3	(20)	
2%		DB	23.6	26.6	29.4	31.5	32.2	31.1	30.0	30.3	30.2	28.9	26.1	23.4	(21)	
		MCWB	13.0	14.0	15.2	17.1	19.0	21.2	22.4	22.5	21.7	20.0	17.1	14.7	(22)	
5%		DB	22.5	25.3	28.4	30.5	30.9	29.6	28.7	29.3	29.1	27.8	25.0	22.3	(23)	
		MCWB	12.9	13.6	15.1	17.0	19.1	21.4	22.3	22.4	21.5	19.6	16.8	14.3	(24)	
10%		DB	21.2	24.0	27.1	29.2	29.5	28.2	27.4	28.0	27.9	26.6	23.7	21.0	(25)	
		MCWB	12.5	13.2	14.8	16.9	19.3	21.5	22.1	22.2	21.4	19.5	16.4	13.9	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.5	16.1	17.3	20.0	22.2	23.2	23.5	23.7	23.1	21.9	20.1	17.6	(27)
MCDWB			20.9	24.3	25.2	26.8	26.5	27.8	28.5	29.2	28.3	27.3	23.7	20.7	(28)	
2%		WB	14.5	15.2	16.6	19.1	21.6	22.7	23.1	23.2	22.6	21.3	19.2	16.5	(29)	
		MCDWB	20.6	23.6	25.1	25.9	25.8	26.9	27.6	28.1	27.5	26.2	22.7	20.4	(30)	
5%		WB	13.7	14.4	16.1	18.5	21.1	22.5	22.7	22.8	22.2	20.9	18.5	15.5	(31)	
		MCDWB	20.0	22.7	24.9	25.6	25.3	26.4	26.7	27.3	26.8	25.4	21.9	20.0	(32)	
10%		WB	13.0	13.8	15.6	18.0	20.7	22.1	22.4	22.5	21.9	20.5	17.7	14.6	(33)	
		MCDWB	19.5	21.8	24.3	25.4	25.1	25.8	26.0	26.6	26.2	24.4	21.3	19.5	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.1	16.2	15.0	13.1	9.6	6.9	6.1	7.2	8.4	9.7	12.5	14.4	(35)
MCDBR			17.9	18.2	17.0	15.1	12.5	9.7	8.6	9.5	10.3	12.0	14.0	15.8	(36)	
MCWBR			8.0	6.7	5.3	4.0	2.7	2.2	2.2	2.6	2.8	3.7	5.6	7.4	(37)	
5% WB		MCDWB	15.5	15.8	13.9	11.8	8.0	6.8	7.0	8.1	8.6	8.9	10.0	12.6	(38)	
		MCWBR	7.4	6.3	4.7	3.6	2.3	2.0	2.1	2.5	2.6	2.8	4.1	6.2	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.337	0.387	0.613	0.607	0.472	0.424	0.418	0.451	0.442	0.418	0.361	0.340	(40)	
taud		2.403	2.262	1.653	1.736	2.187	2.388	2.393	2.274	2.275	2.306	2.398	2.421	(41)		
Ebn at Noon		926	896	714	731	832	866	871	846	849	854	891	907	(42)		
Edh at Noon		106	131	252	236	150	121	121	136	133	124	106	101	(43)		
(44)	All-Sky Solar Radiation	RadAvg	4.37	5.25	5.42	5.45	4.92	4.20	3.84	4.20	4.23	4.05	4.21	3.90	(44)	
(45)		RadStd	0.32	0.39	0.40	0.43	0.52	0.41	0.26	0.33	0.30	0.44	0.33	0.31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.28	+0.34	N/A	+0.51	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+0.28	N/A	N/A	+0.42	N/A	N/A	N/A	-34	+114	+81

Nomenclature: See separate page